

Questions

Montague Hamilton

2 October 2026

1 A - logs and exponentials

1. Evaluate:

$$\log_5(125) = 3$$

$$\log_4\left(\frac{1}{16}\right) = -4$$

$$\ln(e^2) + e^{\ln(5)} = 2 + 5 = 7$$

2. Expand using the log laws:

$$\log_b\left(\frac{x^3}{y^2}\right) = \log_b(x^3) - \log_b(y^2)$$

3. Solve, showing lines:

$$3^{x+1} = 27$$

$$\log_3(27) = x + 1$$

$$x = 2$$

2 B - proofs

4. Prove that 2^x is injective. Name the technique that you are using:

Proof (direct): Suppose,

$$2^a = 2^b$$

$$\ln(2^a) = \ln(2^b)$$

$$a \ln(2) = b \ln(2)$$

$$a = b$$

Because $2^a = 2^b$ leads to $a = b$ the function 2^x is injective.

3 C - transformations

5. **Estimate $\sin(0.03)$ without a calculator, and explain why in one sentence why $\sin \theta < \theta$** At low values of sine, most of the change in arc length of the circle is in height, so $\sin(0.03) \approx 0.03$.
6. **A wheel has centre height 5, radius 2, one full turn every 10 seconds, and the rider starts at the midline moving upward. Write $h(t)$, then find the height at $t = 2.5$ s.**

$$h = 6 \quad r = 2 \quad P = 10$$

At $t = 2.5$

$$\begin{aligned} h(t) &= 2 \sin\left(\frac{2\pi t}{10}\right) + 6 \\ &= 2 \sin\left(\frac{\pi}{2}\right) + 6 \\ &= 2 + 6 \\ &= 8 \end{aligned}$$